FALCON: Ontology-Based Reasoning for Food Allergen and Label Compliance Across Three Regulatory Markets

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Abstract

Undeclared allergens remain a persistent cause of packaged-food recalls, and the rules that decide what a label must declare differ between markets. This report presents FALCON, a knowledge representation system that reasons about allergens, diet claims, and labelling duties for products sold in the United States, Canada, and the European Union. The domain is modelled as an OWL 2 DL ontology in SROIQ(D), built with owlready2: 92 classes, 33 properties, 4 property chains, and 262 individuals describing 45 products and 149 ingredients. Seventeen SWRL rules run in the Pellet reasoner and in a purpose-built engine that adds forward and backward chaining with proof trees, negation as failure under a local closed-world assumption, and defeasible regulatory exemptions. The system answers 14 competency questions. On the curated knowledge base it contradicted 10 of 45 label claims, found 17 undeclared allergen obligations across 10 products, and exposed 10 hidden allergens. A naive closed-world reading would have issued 23 false assurances (19 on safety, 4 on vegan status), and keyword screening reached a precision of only 0.82. Pellet agreed with the engine on all 888 facts it inferred but needed about 35 minutes, against 0.13 seconds.

Keywords: knowledge representation, OWL 2, SWRL, food allergens, label compliance, closed-world assumption

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Introduction

About one in ten adults in the United States reports a food allergy with convincing symptoms (Gupta et al., 2019), and for these consumers the ingredient list is the main safety control. When that control fails, the usual result is a recall: after the Food Allergen Labeling and Consumer Protection Act took effect, undeclared allergens, most often milk, remained a persistent cause of FDA allergen recalls (Gendel & Zhu, 2013).

Part of the difficulty is that allergen knowledge is relational. Whey is milk, Worcestershire sauce is traditionally made with anchovies, and the lysozyme that preserves some hard cheeses comes from egg white, yet none of this is visible in the ingredient name. The rules are also jurisdictional. Pine nuts are a declarable tree nut in the United States and Canada but not in the European Union; mustard, celery, and lupin are regulated in the EU but not by the FDA; and exactly 10 mg/kg of sulphites requires a declaration in North America but not in Europe (European Parliament & Council of the European Union, 2011; Food: Exemptions From Labeling, 2025; Health Canada, n.d.; U.S. Food and Drug Administration [FDA], 2025).

This makes labelling a natural fit for knowledge representation. The concepts form hierarchies, the relationships chain (an ingredient keeps the allergens of whatever it was made from), and the regulations read as rules with thresholds and exceptions. FALCON, the Food Allergen and Label Compliance Ontology, encodes this knowledge and reasons over it.

Scope and Boundaries

FALCON covers packaged foods sold in three markets, 28 allergens (15 regulated groups and 13 named members such as individual tree nuts), ingredient provenance and composition, 10

label claims, and 9 consumer profiles. It leaves out nutrition labelling, the probability of cross-contact, and quantitative thresholds other than those for gluten and sulphites. The knowledge base is curated rather than scraped: each of its 45 products exists to exercise a particular question or edge case, so the counts reported here describe what the system can detect, not how often problems occur in the market.

Competency Questions

Following Grüninger and Fox (1995), the requirements were written as questions the system must answer. The 14 questions range from lookups ("which ingredients are made from milk?") through rule-based inference (which claims are false, and which allergens are undeclared in which market) to questions that need negation ("which products can be confirmed safe for this consumer?") and the classification of a product the system has never seen. They are listed with their answers in the Evaluation section.

Background and Literature Review

An ontology is an explicit specification of a shared conceptualization (Gruber, 1993). In the food domain the most developed example is FoodOn, which harmonizes food sources, processes, and products under the OBO Foundry principles (Dooley et al., 2018; Smith et al., 2007). FoodOn describes what foods are and how they are made, but it does not encode labelling duties. FoodKG links recipes, ingredients, and nutrient data in a knowledge graph for food recommendation (Haussmann et al., 2019); it treats ingredients as graph nodes, as FALCON does, but answers questions by retrieval rather than inference. Neither resource represents market-specific allergen law, which is the gap FALCON fills. Its ingredient classes follow conventions that would allow later alignment with FoodOn through MIREOT (Courtot et al.,

2011), and its data files are documented and machine-readable in line with the FAIR principles (Wilkinson et al., 2016).

OWL 2 DL rests on the description logic SROIQ(D) (Horrocks et al., 2006). It provides the hierarchies, defined classes, and property chains this domain needs, and reasoners such as HermiT (Glimm et al., 2014) and Pellet (Sirin et al., 2007) classify ontologies soundly and completely. Rules that OWL cannot express are added with SWRL (Horrocks et al., 2004), and decidability is kept by applying rules only to named individuals, the DL-safety condition (Motik et al., 2005).

Two features of the domain fall outside both formalisms. OWL makes the open-world assumption, so the absence of an allergen from the data never proves that a product is free of it (Rector et al., 2004). Regulatory exemptions are defeasible: a duty holds unless an exception applies, and a monotonic logic cannot withdraw a conclusion once drawn (Antoniou et al., 2001). Logic programming handles both through negation as failure under the closed-world assumption (Reiter, 1978), with a clear semantics for stratified programs (Apt et al., 1988), and the overlap between description logics and Horn rules is well understood (Grosof et al., 2003).

Choice of Methodology

I therefore chose a hybrid design, developed iteratively from the competency questions as Noy and McGuinness (2001) recommend. The ontology, built with owlready2 (Lamy, 2017), carries the terminology and the monotonic rules and is checked by both reasoners. A separate Datalog-style engine runs the same SWRL rules and adds negation as failure, but only under a local closed-world assumption (Etzioni et al., 1997): closed-world conclusions are drawn only for records marked complete. Running the same rules in two independent engines gives each an

external check. Claude (Anthropic, 2026) assisted with design discussions, code drafting, debugging, and editing; every source, rule, and result was checked independently.

Methodology and System Design

Domain Analysis

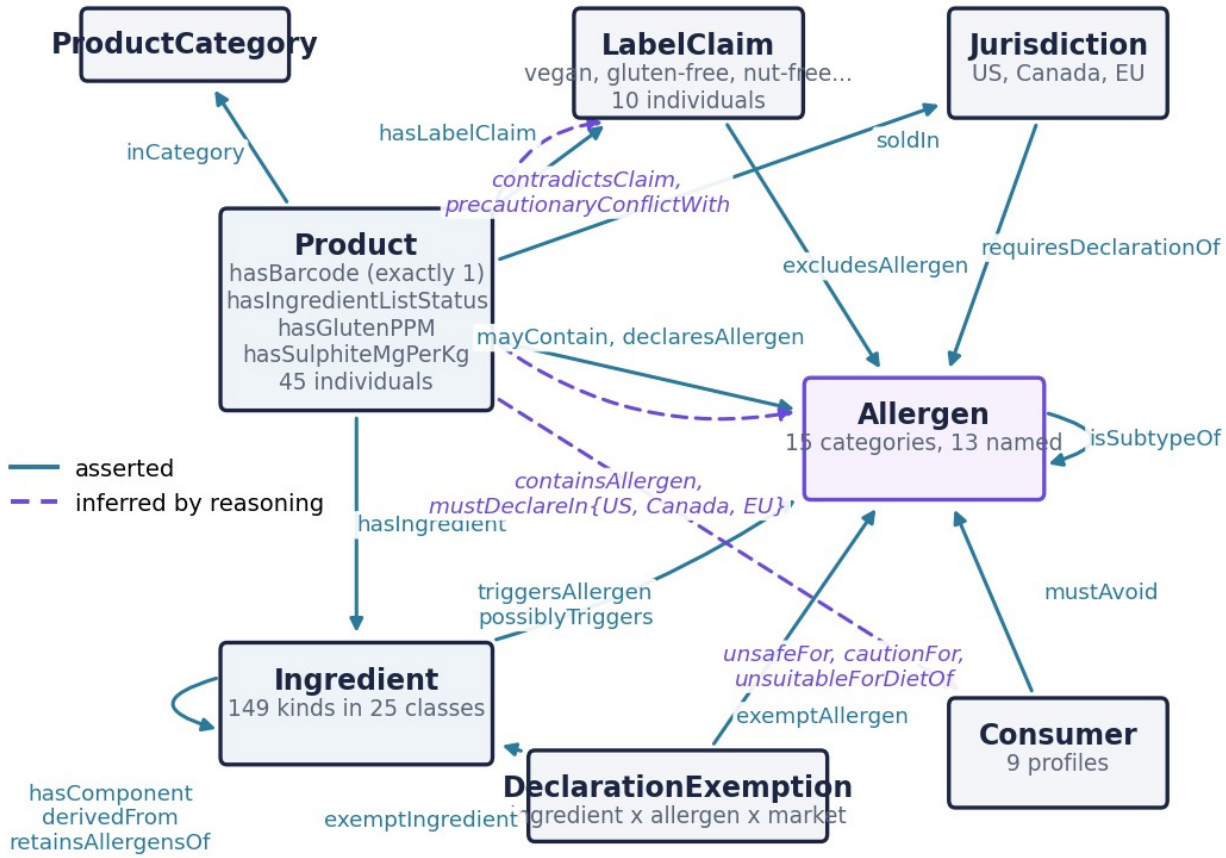
The domain has six kinds of entity: products, ingredient kinds, allergens, label claims, markets, and consumers. Their relationships come from three bodies of knowledge. Ingredient knowledge records what an ingredient is made from and whether processing keeps the source's proteins: whey keeps those of milk, while fully refined soybean oil, though made from soybeans, does not. Regulatory knowledge records which allergens each market requires to be declared, with thresholds and exemptions. Claim knowledge records what each claim rules out, and here the domain is full of traps. "Non-dairy" is a US regulatory term that permits caseinate, a milk protein (Food; Designation of Ingredients, 2025; Food Allergy Research & Resource Program [FARRP], n.d.); "lactose-free" says nothing about milk protein; and "gluten-free" does not mean wheat-free when Codex wheat starch is used (Codex Alimentarius Commission, 2008).

Conceptual Model

Figure 1 shows the core concepts. Solid edges are asserted in the data and dashed edges are inferred. The product is the hub: everything the system concludes about it, from its allergens to the status of its claims, its duties in each market, and its safety for each consumer, is derived from its ingredients, its claims, and where it is sold.

Figure 1

Conceptual Model of FALCON



Note. Boxes are top-level classes; edges are object properties. Dashed edges are never asserted and are produced only by reasoning.

Knowledge Representation Framework

Option C, Python with owlready2, is the primary implementation. The ontology is generated from one Python specification, so it can be rebuilt and tested on every change; owlready2 runs HermiT and Pellet, executes SPARQL, and saves RDF/XML that opens in Protégé 5; and because the specification is ordinary Python, it can also drive a second reasoner. A Protégé-only build would have produced the same ontology without automated tests, and the

OBO-format tools (pronto, OAK) lack the property chains and universal restrictions the model depends on.

Ontology Design Patterns

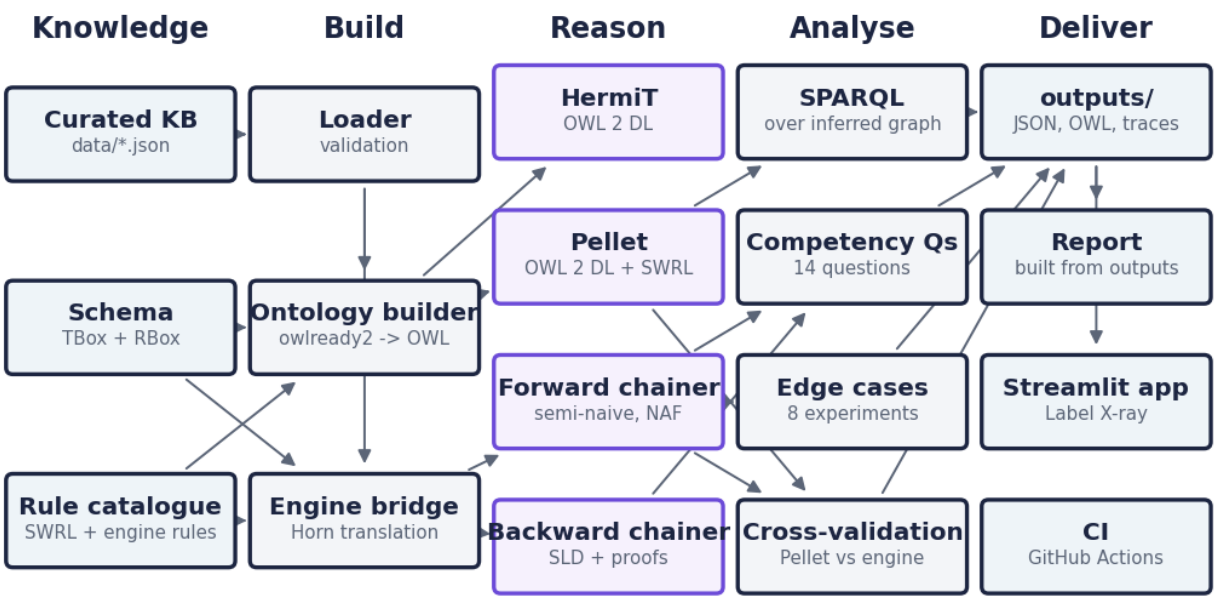
Six patterns shaped the model. Ingredient kinds are individuals rather than classes, because SWRL rules bind only named individuals. Provenance and allergen retention are separate properties: retainsAllergensOf is a sub-property of derivedFrom, and only it carries allergens through a property chain, which reproduces the refined-oil exemption without an exception rule. Gluten is a hazard separate from the wheat allergen, so a coeliac consumer and a wheat-allergic consumer get different answers for the same loaf. The three-way fact "this ingredient is exempt from declaring this allergen in this market" uses the W3C n-ary relation pattern (Noy & Rector, 2006). Label claims are data rather than axioms, so a false claim is reported instead of making the ontology inconsistent. And allergen-group membership is recorded per market, because the markets disagree about what counts as a tree nut.

System Design Principles

Four engineering principles run through the system (Figure 2). One schema module is the single source of truth for both the OWL builder and the rule engine. The knowledge base is validated on load for dangling references and circular derivations. Conclusions are explainable by construction, because the engine stores the premises of every fact it derives. And the pipeline is reproducible: one command regenerates every output, 31 automated tests pin down the behaviour this report relies on, and a continuous-integration workflow repeats the full run, Pellet and HermiT included, on every commit (the Pellet bundled with owlready2 0.51 needs Java 25, which the setup installs).

Figure 2

System Architecture



Note. Arrows show data flow. The same schema and rule catalogue feed both the OWL ontology and the rule engine, and every figure and table in this report is generated from the outputs folder.

Implementation

Ontology Structure

Table 1 summarizes the ontology. Its class hierarchy (Figure 3) has four origin branches for ingredients (plant, animal, other, and compound), with VegetarianIncompatibleIngredient as a second parent for fish, shellfish, meat, slaughter by-products, and insect-derived additives. Defined classes do most of the classification. For example, NonVeganIngredient ≡ AnimalDerivedIngredient ⊔ ∃hasComponent.AnimalDerivedIngredient, and because hasComponent is transitive, an anchovy two levels inside a compound ingredient is enough. Four property chains propagate allergens: retainsAllergensOf ∘ triggersAllergen ⊑ triggersAllergen

and hasComponent ∘ triggersAllergen ⊑ triggersAllergen move them up through ingredients, and two further chains lift containsAllergen and mayContain through the allergen hierarchy. A general class axiom, ∃derivedFrom.AnimalDerivedIngredient ⊑ AnimalDerivedIngredient, passes animal origin through any derivation. Restrictions include an exact cardinality (every product has exactly one barcode) and a qualified one, MultiMarketProduct ≡ Product ⊓ ≥2 soldIn.Jurisdiction, which depends on AllDifferent assertions because OWL makes no unique-name assumption.

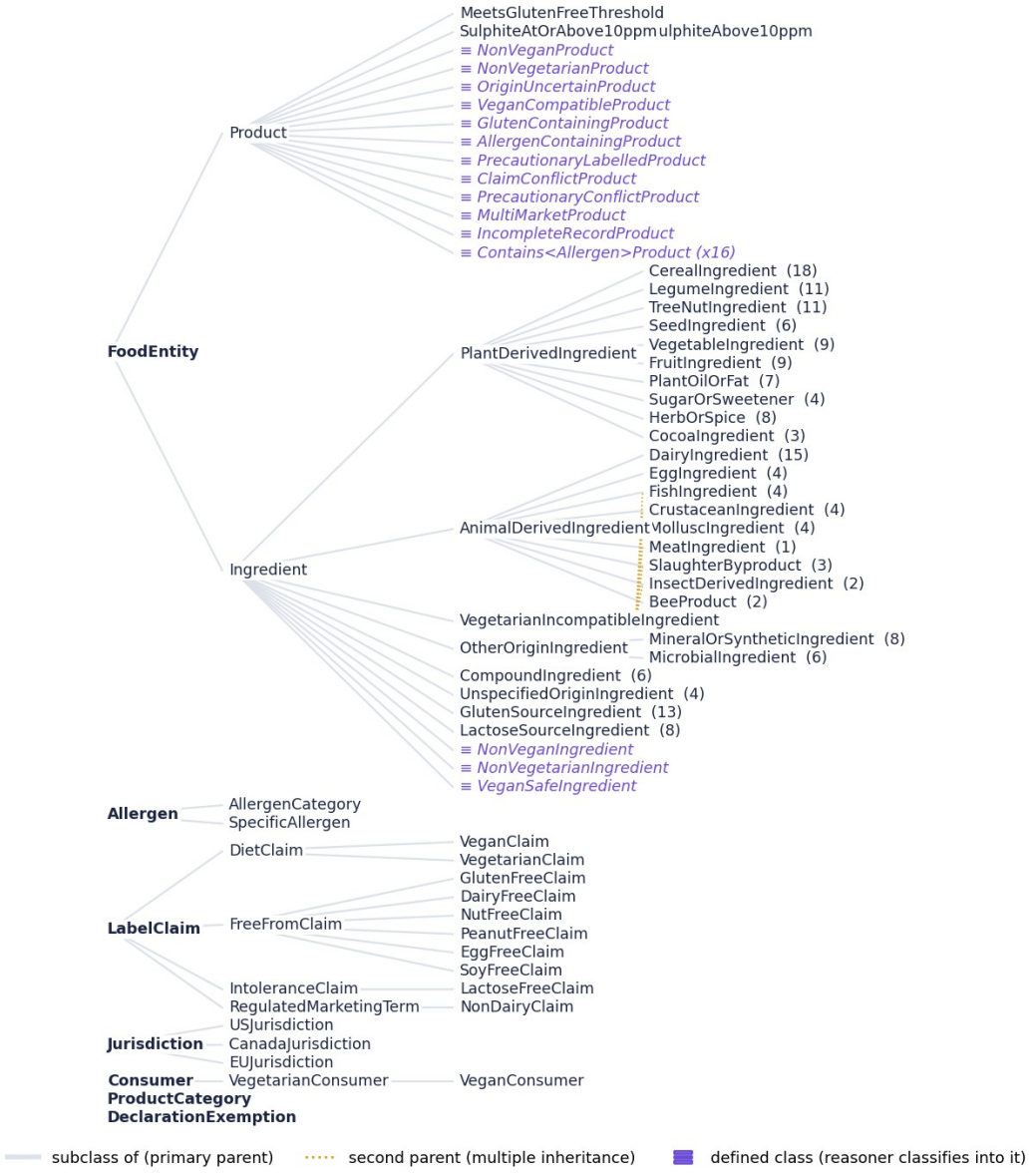
Table 1

Ontology Metrics

Element	Count	Notes
Named classes	62	Asserted hierarchy, including two classes with multiple parents
Defined classes (owl:equivalentClass)	30	16 allergen-specific product classes plus 14 others
Object properties	29	4 transitive, 4 sub-properties, 1 inverse pair
Data properties	4	All functional
Property chains	4	Allergen propagation through ingredients and allergen groups
General class axioms	3	Origin and gluten/lactose inheritance
Cardinality and value restrictions	8	Exact, minimum, maximum and hasValue
Disjointness axioms	6	Top-level kinds, ingredient origins, animal sources, markets, claims
Individuals	262	45 products, 149 ingredients, 28 allergens and reference data
SWRL rules	17	4 use comparison built-ins
Expressivity	SROIQ(D)	Full OWL 2 DL

Figure 3

Class Hierarchy



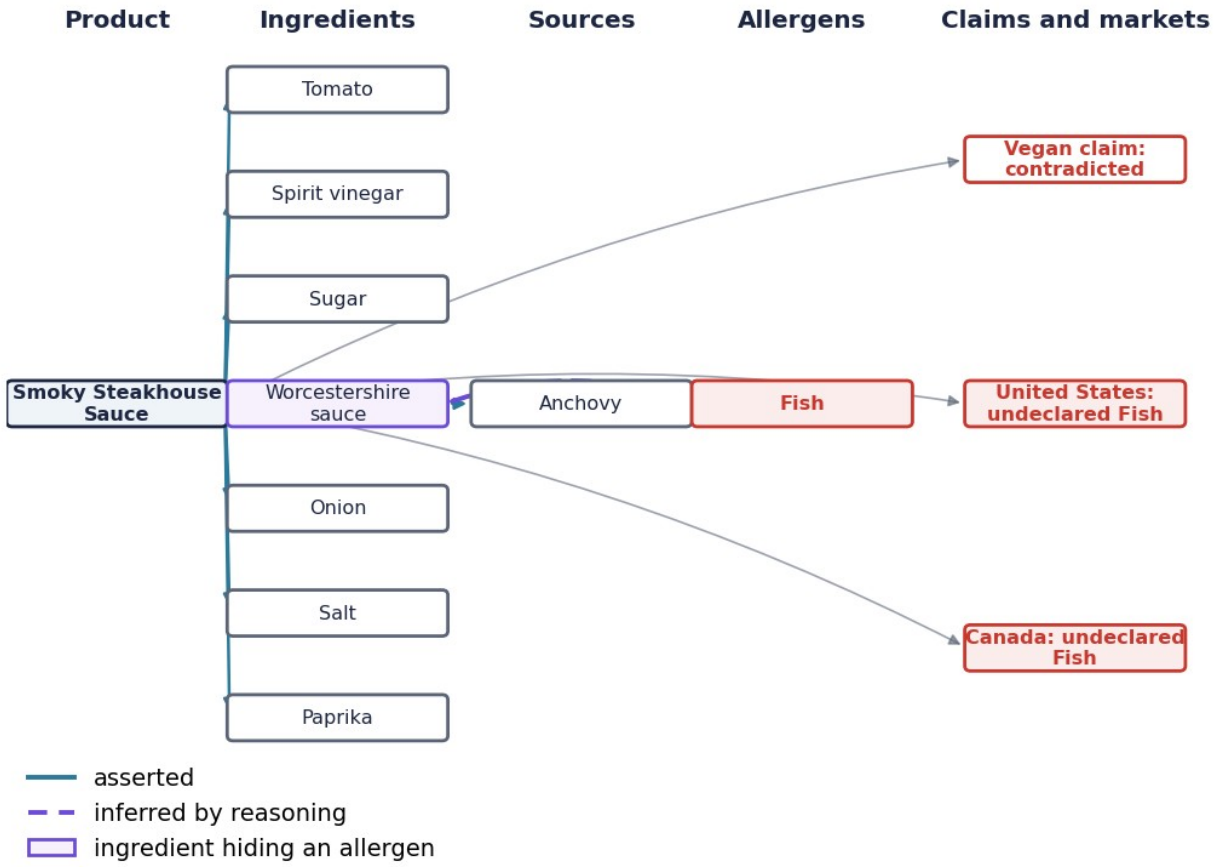
Note. Numbers in parentheses count asserted ingredient individuals. Classes marked ≡ are defined classes that the reasoner populates; dotted lines show second parents.

Figure 4 shows what this structure does for one product. Smoky Steakhouse Sauce lists seven ingredients and claims to be vegan. Nothing on its label says fish, but the reasoner follows

Worcestershire sauce to its anchovy component, concludes that the sauce contains fish and is not vegan, and finds that both of its markets require fish to be declared.

Figure 4

Knowledge Graph Around One Product



Note. Solid arrows are asserted; dashed arrows are inferred. The highlighted ingredient hides an allergen that its name does not reveal.

Rule Design

The rule base has three layers (Table 2). 17 SWRL production rules cover allergen detection, claim checking, consumer safety, declaration duties, and numeric thresholds. 39 engine-only rules add what SWRL cannot express: three-place relations linking an allergen, the ingredient that carries it, and the market that cares; defeasible duties that an exemption can

switch off; and negation as failure for verdicts such as "confirmed safe." 10 integrity rules treat selected OWL axioms as constraints, the job SHACL would do in production (Knublauch & Kontokostas, 2017). The threshold rules show why exact formulation matters: the EU rule tests greaterThan(x, 10) where the North American rule tests greaterThanOrEqual(x, 10), and a product at exactly 10 mg/kg falls between them.

Table 2

Selected Rules in First-Order Form

ID	Rule	Purpose
R01	∀a, i, p: Product(p) ∧ hasIngredient(p, i) ∧ triggersAllergen(i, a) → containsAllergen(p, a)	Product allergen detection
R02	∀a, c, p: hasLabelClaim(p, c) ∧ excludesAllergen(c, a) ∧ containsAllergen(p, a) → contradictsClaim(p, c)	Free-from claim contradiction
R03	∀c, p: hasLabelClaim(p, c) ∧ VeganClaim(c) ∧ NonVeganProduct(p) → contradictsClaim(p, c)	Vegan claim contradiction
R07	∀a, p, u: Consumer(u) ∧ mustAvoid(u, a) ∧ containsAllergen(p, a) → unsafeFor(p, u)	Unsafe for consumer
R13	∀a, j, p: containsAllergen(p, a) ∧ soldIn(p, j) ∧ EUJurisdiction(j) ∧ requiresDeclarationOf(j, a) → mustDeclareInEU(p, a)	EU declaration duty
R14	∀c, p, x: hasLabelClaim(p, c) ∧ GlutenFreeClaim(c) ∧ hasGlutenPPM(p, x) ∧ x ≥ 20.0 → contradictsClaim(p, c)	Gluten-free threshold exceeded
R17	∀p, x: Product(p) ∧ hasSulphiteMgPerKg(p, x) ∧ x > 10.0 → SulphiteAbove10ppm(p)	Sulphites above 10 mg/kg
E04	∀a, i, j, p: containsVia(p, a, i) ∧ soldIn(p, j) ∧ requiresDeclarationOf(j, a) ∧ not exemptFromDeclaration(i, a, j) → declarationDuty(p, a, j)	Declaration duty unless exempt
E08	∀a, j, p: declarationDuty(p, a, j) ∧ not declaresAllergen(p, a) → undeclaredAllergen(p, a, j)	Undeclared allergen
E10	∀a, i, p: Product(p) ∧ hasIngredient(p, i) ∧ triggersAllergen(i, a) ∧ not assertedTrigger(i, a) ∧ not mentionsAllergen(i, a) → hiddenAllergenVia(p, a, i)	Hidden allergen
E13	∀p, u: Consumer(u) ∧ Product(p) ∧ hasIngredientListStatus(p, complete) ∧ hasSomeIngredient(p) ∧ not unsafeFor(p, u) ∧ not cautionFor(p, u) ∧ not uncertainFor(p, u) → confirmedSafeFor(p, u)	Confirmed safe for consumer
E21	∀p: Product(p) ∧ hasIngredientListStatus(p, complete) ∧ hasSomeIngredient(p) ∧ not hasNonVeganSafeIngredient(p) → VerifiedVeganProduct(p)	Verified vegan product

Note. R rules are SWRL and run in both Pellet and the engine; E rules run in the engine only. "not" denotes negation as failure. The full SWRL catalogue is in Appendix B.

Technical Architecture

The loader validates the JSON knowledge base (Figure 2). The ontology builder turns the schema and data into OWL, and the engine bridge translates the Horn-expressible part of the same schema (sub-class and sub-property axioms, inverses, transitivity, chains, and the sufficient conditions of defined classes, 205 rules in all) into engine rules, which is essentially the OWL 2 RL reading of the axioms (Motik et al., 2012). The forward chainer uses semi-naive evaluation over stratified rules. The backward chainer uses SLD resolution with loop checking and, when a goal fails, names the first condition that blocked each rule. Competency questions, edge-case experiments, and the cross-validation draw on both reasoners, and their results are written to one folder from which this report's figures and tables, and an interactive Streamlit application, are generated.

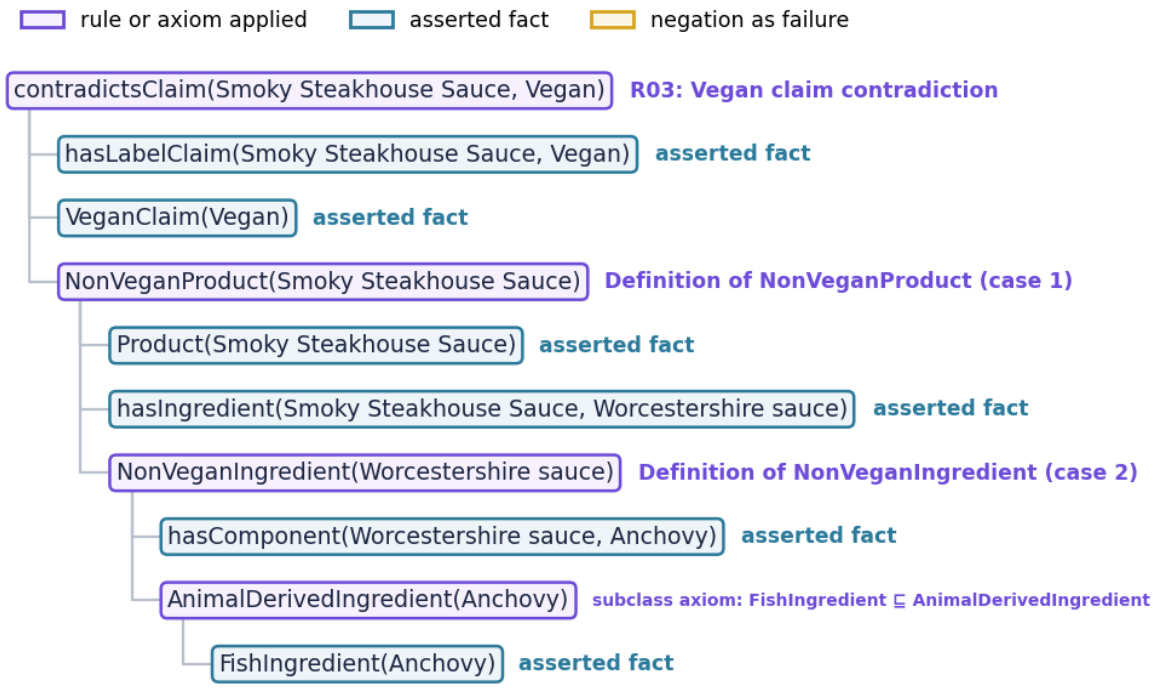
Evaluation and Results

Reasoning Performance

From 1,174 asserted facts, forward chaining derived 2,893 facts across 8 strata in 0.13 seconds. Figure 5 shows the backward-chaining proof that the steakhouse sauce's vegan claim is false; it reaches down through four rule applications to the asserted fact that anchovy is a fish ingredient. On all 8 showcase goals, backward chaining reached the same verdict as forward chaining, in 3.38 ms per goal on average. Table 3 lists each competency question with its answer.

Figure 5

Backward-Chaining Proof That the Steakhouse Sauce's Vegan Claim Is False



Note. Each box is a goal; violet boxes name the rule applied and blue boxes are asserted facts.

The same derivation is recorded, in reverse order, by the forward chainer.

Table 3

Competency Questions and Answers

ID	Question	Answer
CQ01	Which ingredients are made from milk, directly or indirectly?	9 ingredients asserted as made directly from milk; reasoning finds 14 in total.
CQ02	Which allergens does Smoky Steakhouse Sauce contain, and through which ingredient?	Fish via Worcestershire sauce (hidden)
CQ03	Which products contradict their own label claims, and why?	10 of 45 claims are contradicted by the products' own composition.
CQ04	Which free-from claims are undermined by a 'may contain' warning?	Dark Chocolate 70%: Dairy-free claim vs. 'may contain milk'; Nut-Free Granola Bar: Nut-free claim vs. 'may contain tree nuts'
CQ05	Which products are unsafe, or need caution, for Liam (peanut and tree nut allergy)?	7 unsafe, 2 caution, 0 uncertain
CQ06	Which products can be confirmed safe for Liam, and why are others excluded?	34 products confirmed safe; Dark Chocolate 70% blocked by its hazelnut warning

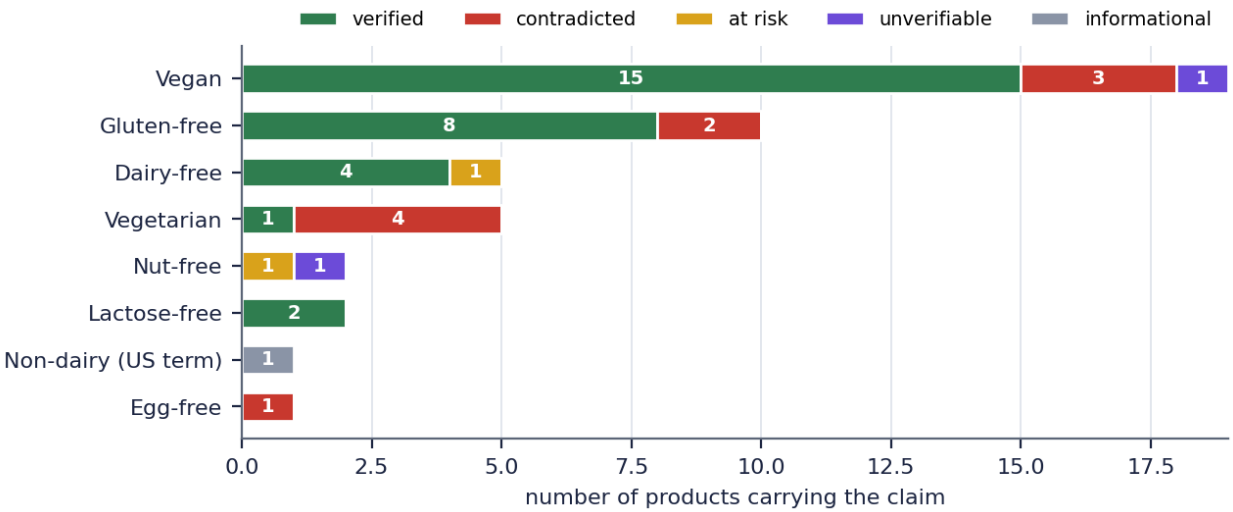
ID	Question	Answer
CQ07	How is each product classified automatically?	45 of 45 products placed in at least one defined class
CQ08	Which allergens must each product declare in the US, Canada and the EU, and where do the markets disagree?	11 product-allergen pairs are treated differently by markets the product is sold in.
CQ09	Which products contain an allergen that the ingredient names do not reveal?	10 hidden allergens in 9 products (e.g., caseinate, tahini, surimi)
CQ10	Which products cannot be assessed because their ingredient information is incomplete?	2 incomplete records; neither can be confirmed safe for any consumer
CQ11	Which gluten-free products meet the 20 ppm limit, which exceed it, and which rely on the recipe alone?	10 claims: 8 verified, 2 contradicted
CQ12	How is a new, previously unseen product classified?	NonVeganProduct; vegan claim contradicted; wheat undeclared ((Canada), (European Union))
CQ13	Which products have allergens that a market requires but the label does not declare?	10 products carry at least one undeclared allergen in a market where they are sold.
CQ14	Which label claims are verified, contradicted, at risk or unverifiable?	30 verified, 10 contradicted, 2 at risk, 2 unverifiable, 1 informational

Note. Answers come from the rule engine; SPARQL versions of CQ01 to CQ11, CQ13, and CQ14 run against the Pellet-materialized ontology.

The claim audit is the clearest result (Figure 6). Of 45 claims, 30 were verified, 10 were contradicted by the product's own composition, 2 were undermined by a "may contain" warning, and 2 could be neither confirmed nor refuted; the US term "non-dairy" is informational only. Contradictions arrived through every route the model encodes: insects (carmine in "vegan" gummies), honey, animal rennet inside parmesan, barley malt in "gluten-free" flakes, and egg yolk inside the mayonnaise of an "egg-free" coleslaw. On compliance, 17 undeclared allergen obligations appeared in 10 products, and 11 product–allergen pairs were treated differently by markets that sell the same product (Figure 7). Figure 8 shows the consequence for each of the nine consumer profiles.

Figure 6

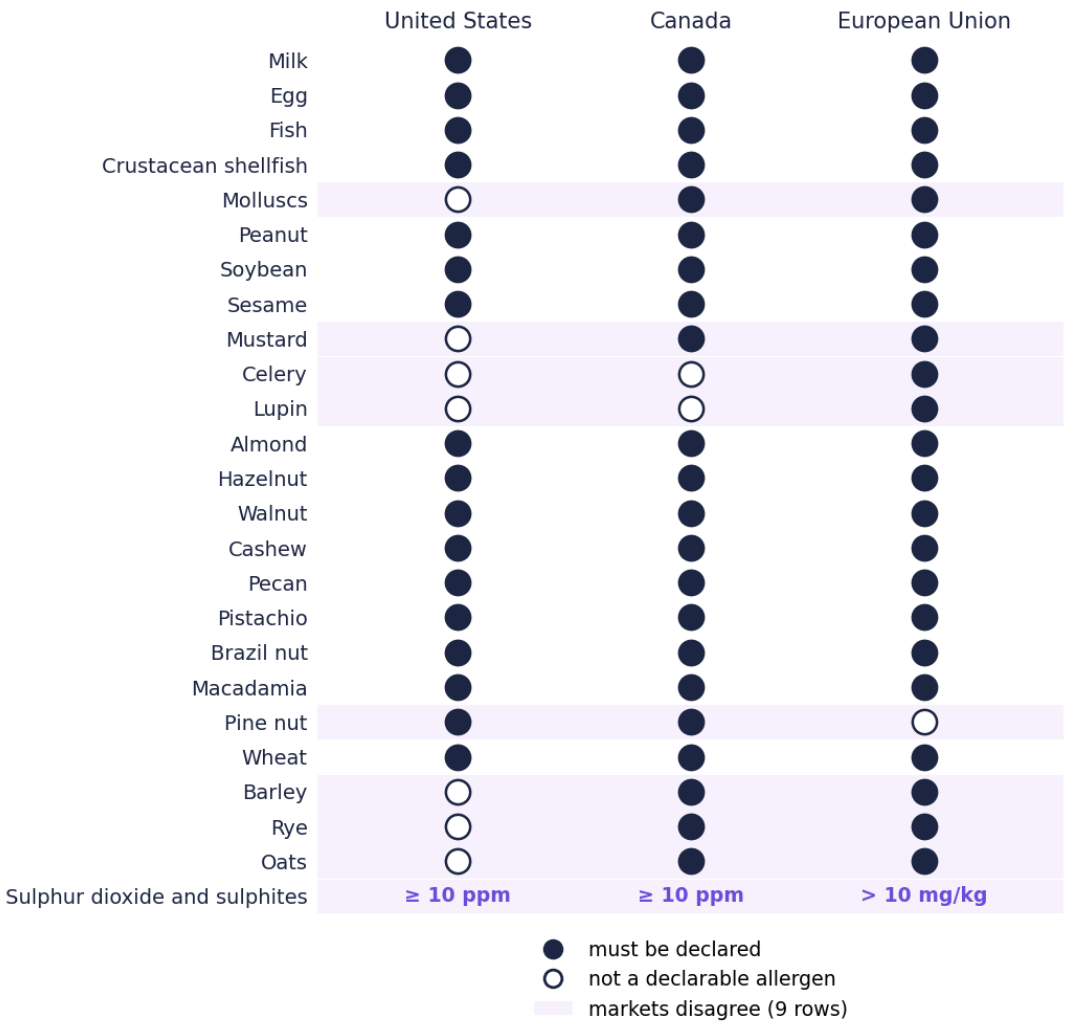
Audit of Every Label Claim in the Knowledge Base



Note. Verified claims hold under the local closed-world assumption; unverifiable claims are not contradicted but depend on missing or ambiguous data.

Figure 7

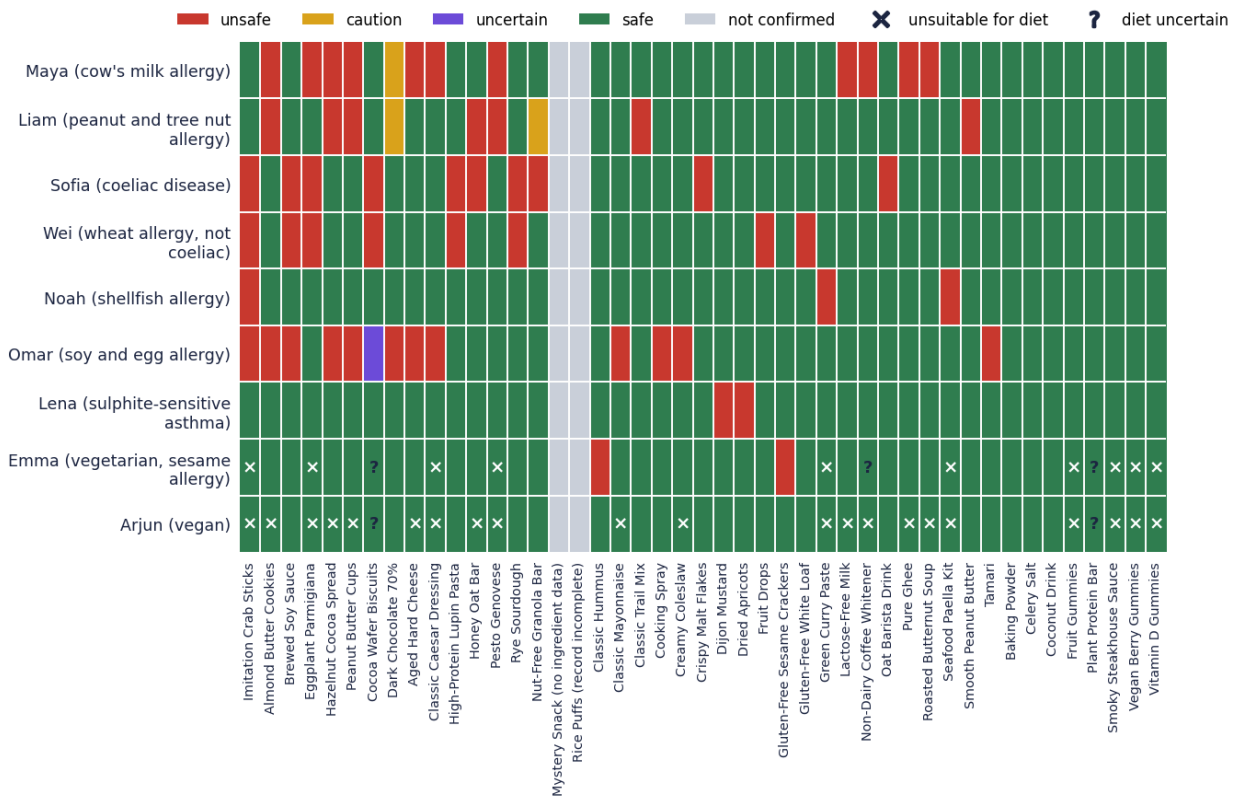
Allergens Each Market Requires to Be Declared



Note. Shaded rows are where the three markets disagree. Sulphite thresholds differ in their comparison operator, not their value.

Figure 8

Safety Verdict for Every Product and Consumer Profile



Note. Grey columns are the two incomplete records, which can never be confirmed safe. A cross marks a diet the product does not suit.

Consistency and Completeness

Pellet reasoned over the complete ontology, including all 17 SWRL rules, found it consistent, and inferred 888 facts; materialising them took 2,075 seconds (about 35 minutes). HermiT, given the 13 rules without built-ins, reported "consistent" in 5.8 seconds. Neither found an unsatisfiable class. Cross-validation compared 47 predicates and found all 888 facts in agreement between Pellet and the rule engine, which derives them in 0.13 seconds. Table 4 compares the reasoner runs. Within the engine, the integrity rules found no violations in the curated data, and all 31 tests pass. Completeness was judged against the requirements: every

competency question is answered, every edge case behaves as expected, and every product exercises at least one rule. The engine's closed-world classes are deliberately not derivable in OWL; under the open-world assumption VeganCompatibleProduct has no members; once closure axioms were added to the complete records, Pellet classified exactly the 20 products as vegan-compatible that the engine verifies.

Table 4

Reasoner Runs on the Curated Knowledge Base

Run	Result	Time
FALCON engine, forward chaining	2,893 facts derived, 8 strata	0.13 s
FALCON engine, backward chaining	8 showcase goals, all agree with forward	3.38 ms per goal
Pellet (OWL 2 DL + 17 SWRL rules)	consistent; 888 facts inferred	2,075 s (34.6 min)
HermiT (OWL 2 DL + DL-safe rules)	consistent	5.80 s
Pellet vs engine cross-validation	888 of 888 facts agree; 47 predicates compared	

Note. All figures come from one complete run on Google Colab (Python 3.13, owlready2 0.51, Java 25); see outputs/owl_results.json.

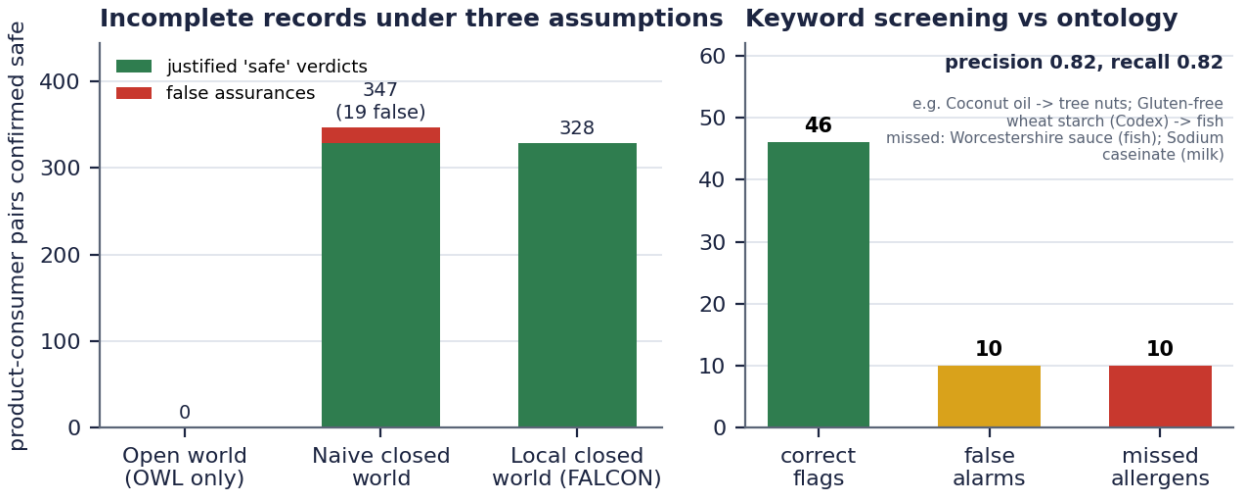
Edge Cases

Incomplete information was tested under three assumptions (Figure 9, left). Under the open-world assumption no product could be confirmed safe or vegan. A naive closed-world reading, in which absence of evidence counts as safety, confirmed 347 product–consumer pairs, 19 of them wrongly, and gave 4 false vegan verdicts, all from the two incomplete records or from ingredients of unstated origin. The local closed-world assumption confirmed 328 pairs with no false assurances. Conflicting data behaved as designed: asserting vegan claims as axioms makes the ontology inconsistent (Pellet detected the inconsistency in 0.9 seconds), whereas the engine reports each contradiction and keeps every unrelated conclusion. An injected error, honey typed as plant-derived, was caught by a disjointness rule while all 55 unsafe verdicts stayed

unchanged; in the ontology the same error is a contradiction, which Pellet reported in 0.8 seconds. Unspecified lecithin produced "uncertain" verdicts and 4 supplier checks rather than guesses. Finally, a keyword screener reached a precision and recall of 0.82 (Figure 9, right). It flagged cocoa butter as dairy and Codex wheat starch as fish (because of the letters "cod"), and it missed caseinate, tahini, and surimi.

Figure 9

Incomplete Records Under Three Assumptions, and Keyword Screening Versus the Ontology



Note. Left: product–consumer pairs confirmed safe; red marks false assurances. Right: allergen detections by substring matching, scored against the ontology's conclusions.

Limitations Identified

The system cannot represent ingredient quantities, so percentage thresholds and dose are out of reach. It has no model of cross-contact probability, so precautionary labels are taken at face value. It has no notion of time, so a recipe change or regulatory update overwrites the old knowledge instead of coexisting with it. Three-place relations exist in OWL only as reified individuals, SWRL can neither create individuals nor retract conclusions, and the regulatory lists are simplified; Canada's rule that directly added sulphites must be declared at any level, for

example, is not modelled. The knowledge base is small, and matching free-text labels relies on a hand-built lexicon.

Critical Analysis

Expressiveness Versus Computational Complexity

The ontology uses intersection, union, complement, existential and universal restrictions, a property hierarchy, transitive and inverse properties, nominals, qualified cardinality, property chains, and datatypes, which places it in SROIQ(D), the full OWL 2 DL. Consistency checking there is N2ExpTime-complete in the worst case (Kazakov, 2008). Even this small ontology made the practical cost visible: with hasValue nominals, SWRL rules, and full property-value inference all in play, every complete Pellet run took 25 to 36 minutes, although Pellet detected each injected contradiction in about a second. The engine sits at the other end of the trade-off: Datalog with stratified negation has polynomial data complexity, so its runtime grows predictably with the number of products. The price is expressiveness. The engine sees only the Horn part of the axioms, so a universal restriction such as VeganCompatibleProduct must be rewritten as an explicitly closed-world rule. The OWL 2 RL profile would give polynomial reasoning inside OWL (Motik et al., 2012), but it gives up the universal restrictions and unions on which vegan classification depends.

Alternative Approaches

Three alternatives were weighed. A Protégé-only build (Option A) would have produced the same ontology but no tests, no negation as failure, and no readable proofs, and every edge case in Figure 9 would collapse to the open-world row. A pure logic program would have made exemptions and closed-world verification easier but lost open-world classification and the standard tooling that lets others reuse the ontology. Knowledge-graph embeddings (Option B),

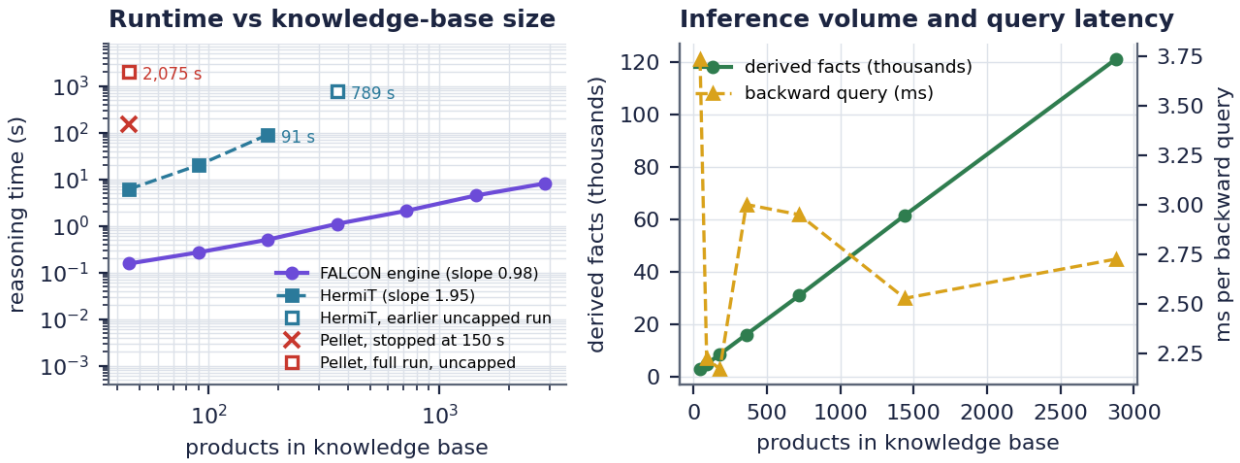
such as TransE (Bordes et al., 2013) trained with pykg2vec (Yu et al., 2021), would learn to rank likely missing links, such as a probable "may contain" warning for a product made on shared lines (Hogan et al., 2021). That suits triage, but it replaces a proof with a score: every result in this report would become a probability, and a hidden allergen would be flagged only if similar products had been labelled the same way. For a safety question that is unacceptable as the primary method, though useful as a secondary one.

Scalability Considerations

The engine was benchmarked on synthetic knowledge bases that replicate the curated products with one random ingredient swapped per copy (Figure 10). From 45 to 2,880 products, runtime grew in proportion to size (fitted exponent 0.976), reaching 8.2 seconds for 121,293 derived facts, while backward queries stayed near 2.7 ms. On the same synthetic data, HermiT needed 6.1, 20.1 and 90.6 seconds at 45, 90 and 180 products (fitted exponent 1.945), and an earlier uncapped run took 789 seconds at 360 products. Pellet could not finish even the 45-product knowledge base within the 150-second limit; its full run on the curated data took 2,075 seconds (about 35 minutes), against 0.13 seconds for the engine. Because products are independent of one another, a production system could classify the shared terminology once and partition the product records, keeping description-logic reasoning for the ontology and a Datalog or OWL 2 RL engine for the millions of records in open food databases.

Figure 10

Scalability of Reasoning With Knowledge-Base Size



Note. Left: wall-clock time on log–log axes, where a slope of 1 means linear growth. Hollow markers are uncapped runs; a cross marks a run stopped at the 150-second limit. Right: derived facts and the median time of a backward-chaining query.

Conclusion

The conclusion is written as a briefing to the head of a food company's regulatory team.

To: Director of Food Safety and Regulatory Affairs

From: Knowledge Engineering, FALCON project

Re: What the allergen knowledge model tells us about labelling risk

Market rules differ in ways that matter. Pine nuts must be declared as a tree nut in the United States and Canada but not in the EU; mustard is declarable in Canada and the EU but not the United States; celery and lupin are EU-only; and at exactly 10 mg/kg of sulphites a product needs a declaration in North America but not in Europe. In our test portfolio, 11 product–allergen pairs were handled differently by markets selling the same product. A single global checklist cannot catch these; compliance has to be checked market by market.

Hidden allergens arrive through ingredients made from, or containing, other ingredients: caseinate in a "non-dairy" whitener, anchovy in Worcestershire sauce, egg-white lysozyme in cheese, tahini in hummus. The model found 10 such cases. Keyword screening both misses these and raises false alarms (precision and recall of 0.82), so ingredient data should be mapped to a structured ontology rather than matched as text.

Several claims do not mean what customers assume. "Non-dairy" permits milk protein, "lactose-free" says nothing about milk allergy, "gluten-free" is not wheat-free, and "dairy-free" next to "may contain milk" sends two opposite messages. Claims and precautionary statements should be reviewed together, against the recipe, before a label is approved.

Data completeness is a safety issue. Treating missing information as good news produced 23 false assurances in our tests. Only records with complete ingredient specifications should receive a "safe" or "vegan" verdict, and ingredients of unstated origin, such as lecithin and E471, need supplier statements; the model raised 4 such requests.

Formal reasoning and fast checking can work together. The OWL reasoner and the rule engine reached the same conclusions on all 888 facts the reasoner inferred, but the reasoner needed about 35 minutes where the engine needed 0.13 seconds. The reasoner should validate the terminology whenever it changes; the engine should run the day-to-day label checks.

These figures come from a curated test portfolio, so they show what the method can detect rather than how often problems occur. The next phase should load real product data from an open food database, align ingredient classes with FoodOn, add SHACL validation at data entry, model cross-contact risk explicitly, and test knowledge-graph embeddings as a triage tool that suggests which products to review first.

References

Anthropic. (2026). *Claude* (Opus 5.5 version) [Large language model]. https://claude.ai

Antoniou, G., Billington, D., Governatori, G., & Maher, M. J. (2001). Representation results for defeasible logic. *ACM Transactions on Computational Logic*, *2*(2), 255–287. https://doi.org/10.1145/371316.371517

Apt, K. R., Blair, H. A., & Walker, A. (1988). Towards a theory of declarative knowledge. In J. Minker (Ed.), *Foundations of deductive databases and logic programming* (pp. 89–148). Morgan Kaufmann.

Bordes, A., Usunier, N., Garcia-Durán, A., Weston, J., & Yakhnenko, O. (2013). Translating embeddings for modeling multi-relational data. In *Advances in Neural Information Processing Systems 26* (pp. 2787–2795). Curran Associates.

Codex Alimentarius Commission. (2008). *Standard for foods for special dietary use for persons intolerant to gluten* (CXS 118-1979, revised 2008). Food and Agriculture Organization of the United Nations & World Health Organization.

Courtot, M., Gibson, F., Lister, A. L., Malone, J., Schober, D., Brinkman, R. R., & Ruttenberg, A. (2011). MIREOT: The minimum information to reference an external ontology term. *Applied Ontology*, *6*(1), 23–33. https://doi.org/10.3233/AO-2011-0087

Dooley, D. M., Griffiths, E. J., Gosal, G. S., Buttigieg, P. L., Hoehndorf, R., Lange, M. C., Schriml, L. M., Brinkman, F. S. L., & Hsiao, W. W. L. (2018). FoodOn: A harmonized food ontology to increase global food traceability, quality control and data integration. *npj Science of Food*, *2*, Article 23. https://doi.org/10.1038/s41538-018-0032-6

Etzioni, O., Golden, K., & Weld, D. S. (1997). Sound and efficient closed-world reasoning for planning. *Artificial Intelligence, 89*(1–2), 113–148. https://doi.org/10.1016/S0004-3702(96)00026-4

European Parliament & Council of the European Union. (2011). Regulation (EU) No 1169/2011 of 25 October 2011 on the provision of food information to consumers. *Official Journal of the European Union, L 304*, 18–63. https://eur-lex.europa.eu/eli/reg/2011/1169/oj

Food Allergy Research & Resource Program. (n.d.). *Dairy free and non-dairy*. University of Nebraska–Lincoln. Retrieved October 27, 2026, from https://farrp.unl.edu/resources/gi-fas/opinion-and-summaries/dairy-free-and-non-dairy

Food; Designation of Ingredients, 21 C.F.R. § 101.4 (2025).

Food: Exemptions From Labeling, 21 C.F.R. § 101.100 (2025).

Gendel, S. M., & Zhu, J. (2013). Analysis of U.S. Food and Drug Administration food allergen recalls after implementation of the Food Allergen Labeling and Consumer Protection Act. *Journal of Food Protection, 76*(11), 1933–1938. https://doi.org/10.4315/0362-028X.JFP-13-096

Glimm, B., Horrocks, I., Motik, B., Stoilos, G., & Wang, Z. (2014). HermiT: An OWL 2 reasoner. *Journal of Automated Reasoning*, *53*(3), 245–269. https://doi.org/10.1007/s10817-014-9305-1

Grosof, B. N., Horrocks, I., Volz, R., & Decker, S. (2003). Description logic programs: Combining logic programs with description logic. In *Proceedings of the 12th International Conference on World Wide Web* (pp. 48–57). ACM. https://doi.org/10.1145/775152.775160

Gruber, T. R. (1993). A translation approach to portable ontology specifications. *Knowledge Acquisition, 5*(2), 199–220. https://doi.org/10.1006/knac.1993.1008

Grüninger, M., & Fox, M. S. (1995). Methodology for the design and evaluation of ontologies. In *Proceedings of the Workshop on Basic Ontological Issues in Knowledge Sharing, IJCAI-95*. Montréal, Canada.

Gupta, R. S., Warren, C. M., Smith, B. M., Jiang, J., Blumenstock, J. A., Davis, M. M., Schleimer, R. P., & Nadeau, K. C. (2019). Prevalence and severity of food allergies among US adults. *JAMA Network Open, 2*(1), Article e185630. https://doi.org/10.1001/jamanetworkopen.2018.5630

Haussmann, S., Seneviratne, O., Chen, Y., Ne'eman, Y., Codella, J., Chen, C.-H., McGuinness, D. L., & Zaki, M. J. (2019). FoodKG: A semantics-driven knowledge graph for food recommendation. In *The Semantic Web – ISWC 2019* (Lecture Notes in Computer Science, Vol. 11779, pp. 146–162). Springer. https://doi.org/10.1007/978-3-030-30796-7_10

Health Canada. (n.d.). *Questions and answers about the new regulations to enhance the labelling of food allergens, gluten and added sulphites*. Government of Canada. Retrieved October 27, 2026, from https://www.canada.ca/en/health-canada/services/food-nutrition/food-labelling/allergen-labelling/questions-answers-about-new-regulations-enhance-labelling-food-allergens-gluten-added-sulphites.html

Hogan, A., Blomqvist, E., Cochez, M., d'Amato, C., de Melo, G., Gutierrez, C., Kirrane, S., Labra Gayo, J. E., Navigli, R., Neumaier, S., Ngonga Ngomo, A.-C., Polleres, A., Rashid, S. M., Rula, A., Schmelzeisen, L., Sequeda, J., Staab, S., & Zimmermann, A.

(2021). Knowledge graphs. *ACM Computing Surveys, 54*(4), Article 71. https://doi.org/10.1145/3447772

Horrocks, I., Kutz, O., & Sattler, U. (2006). The even more irresistible SROIQ. In *Proceedings of the 10th International Conference on Principles of Knowledge Representation and Reasoning* (pp. 57–67). AAAI Press.

Horrocks, I., Patel-Schneider, P. F., Boley, H., Tabet, S., Grosof, B., & Dean, M. (2004). *SWRL: A semantic web rule language combining OWL and RuleML* (W3C Member Submission). World Wide Web Consortium. https://www.w3.org/Submission/SWRL/

Kazakov, Y. (2008). RIQ and SROIQ are harder than SHOIQ. In *Proceedings of the 11th International Conference on Principles of Knowledge Representation and Reasoning* (pp. 274–284). AAAI Press.

Knublauch, H., & Kontokostas, D. (Eds.). (2017). *Shapes constraint language (SHACL)* (W3C Recommendation). World Wide Web Consortium. https://www.w3.org/TR/shacl/

Lamy, J.-B. (2017). Owlready: Ontology-oriented programming in Python with automatic classification and high level constructs for biomedical ontologies. *Artificial Intelligence in Medicine, 80*, 11–28. https://doi.org/10.1016/j.artmed.2017.07.002

Motik, B., Cuenca Grau, B., Horrocks, I., Wu, Z., Fokoue, A., & Lutz, C. (Eds.). (2012). *OWL 2 Web Ontology Language profiles* (2nd ed., W3C Recommendation). World Wide Web Consortium. https://www.w3.org/TR/owl2-profiles/

Motik, B., Sattler, U., & Studer, R. (2005). Query answering for OWL-DL with rules. *Journal of Web Semantics, 3*(1), 41–60. https://doi.org/10.1016/j.websem.2005.05.001

Noy, N., & Rector, A. (2006). *Defining n-ary relations on the Semantic Web* (W3C Working Group Note). World Wide Web Consortium. https://www.w3.org/TR/swbp-n-aryRelations/

Noy, N. F., & McGuinness, D. L. (2001). *Ontology development 101: A guide to creating your first ontology* (Technical Report KSL-01-05). Stanford Knowledge Systems Laboratory.

Rector, A., Drummond, N., Horridge, M., Rogers, J., Knublauch, H., Stevens, R., Wang, H., & Wroe, C. (2004). OWL pizzas: Practical experience of teaching OWL-DL: Common errors and common patterns. In E. Motta, N. R. Shadbolt, A. Stutt, & N. Gibbins (Eds.), *Engineering knowledge in the age of the Semantic Web* (Lecture Notes in Computer Science, Vol. 3257, pp. 63–81). Springer. https://doi.org/10.1007/978-3-540-30202-5_5

Reiter, R. (1978). On closed world data bases. In H. Gallaire & J. Minker (Eds.), *Logic and data bases* (pp. 55–76). Plenum Press.

Sirin, E., Parsia, B., Cuenca Grau, B., Kalyanpur, A., & Katz, Y. (2007). Pellet: A practical OWL-DL reasoner. *Journal of Web Semantics, 5*(2), 51–53. https://doi.org/10.1016/j.websem.2007.03.004

Smith, B., Ashburner, M., Rosse, C., Bard, J., Bug, W., Ceusters, W., Goldberg, L. J., Eilbeck, K., Ireland, A., Mungall, C. J., Leontis, N., Rocca-Serra, P., Ruttenberg, A., Sansone, S.-A., Scheuermann, R. H., Shah, N., Whetzel, P. L., & Lewis, S. (2007). The OBO Foundry: Coordinated evolution of ontologies to support biomedical data integration. *Nature Biotechnology, 25*(11), 1251–1255. https://doi.org/10.1038/nbt1346

U.S. Food and Drug Administration. (2025). *Questions and answers regarding food allergens, including the food allergen labeling requirements of the Federal Food, Drug, and*

Cosmetic Act (Edition 5): Guidance for industry. https://www.fda.gov/regulatory-information/search-fda-guidance-documents

Wilkinson, M. D., Dumontier, M., Aalbersberg, I. J., Appleton, G., Axton, M., Baak, A., Blomberg, N., Boiten, J.-W., da Silva Santos, L. B., Bourne, P. E., Bouwman, J., Brookes, A. J., Clark, T., Crosas, M., Dillo, I., Dumon, O., Edmunds, S., Evelo, C. T., Finkers, R., . . . Mons, B. (2016). The FAIR guiding principles for scientific data management and stewardship. *Scientific Data, 3*, Article 160018. https://doi.org/10.1038/sdata.2016.18

Yu, S.-Y., Rokka Chhetri, S., Canedo, A., Goyal, P., & Al Faruque, M. A. (2021). Pykg2vec: A Python library for knowledge graph embedding. *Journal of Machine Learning Research, 22*(16), 1–6.

Appendix A

Reproducing the Results

All code, data, and outputs are in the project repository. The commands below rebuild every number in this report; the continuous-integration workflow runs the same steps on every commit and publishes the outputs folder as a downloadable artifact.

Table A1

Commands

Step	Command
Install (Windows)	setup_windows.ps1 (creates .venv, installs Java if missing)
Check environment	python check_environment.py
Run the tests	python -m unittest discover -s tests -t .
Full pipeline	python run_all.py
Scalability benchmark	python -m falcon.benchmark --owl
Figures	python -m falcon.viz
Interactive app	streamlit run app/streamlit_app.py
Rebuild this report	python report/prepare_data.py; node report/build_report.js

Appendix B

SWRL Rule Catalogue

Table B1

All SWRL Production Rules

ID	Name	Rule
R01	Product allergen detection	∀a, i, p: Product(p) ∧ hasIngredient(p, i) ∧ triggersAllergen(i, a) → containsAllergen(p, a)
R02	Free-from claim contradiction	∀a, c, p: hasLabelClaim(p, c) ∧ excludesAllergen(c, a) ∧ containsAllergen(p, a) → contradictsClaim(p, c)
R03	Vegan claim contradiction	∀c, p: hasLabelClaim(p, c) ∧ VeganClaim(c) ∧ NonVeganProduct(p) → contradictsClaim(p, c)
R04	Vegetarian claim contradiction	∀c, p: hasLabelClaim(p, c) ∧ VegetarianClaim(c) ∧ NonVegetarianProduct(p) → contradictsClaim(p, c)
R05	Lactose-free claim contradiction	∀c, i, p: hasLabelClaim(p, c) ∧ LactoseFreeClaim(c) ∧ hasIngredient(p, i) ∧ LactoseSourceIngredient(i) → contradictsClaim(p, c)
R06	Precautionary label conflict	∀a, c, p: hasLabelClaim(p, c) ∧ excludesAllergen(c, a) ∧ mayContain(p, a) → precautionaryConflictWith(p, c)
R07	Unsafe for consumer	∀a, p, u: Consumer(u) ∧ mustAvoid(u, a) ∧ containsAllergen(p, a) → unsafeFor(p, u)
R08	Caution for consumer	∀a, p, u: Consumer(u) ∧ mustAvoid(u, a) ∧ mayContain(p, a) → cautionFor(p, u)
R09	Unsuitable for vegan diet	∀p, u: VeganConsumer(u) ∧ NonVeganProduct(p) → unsuitableForDietOf(p, u)
R10	Unsuitable for vegetarian diet	∀p, u: VegetarianConsumer(u) ∧ NonVegetarianProduct(p) → unsuitableForDietOf(p, u)
R11	US declaration duty	∀a, j, p: containsAllergen(p, a) ∧ soldIn(p, j) ∧ USJurisdiction(j) ∧ requiresDeclarationOf(j, a) → mustDeclareInUS(p, a)
R12	Canada declaration duty	∀a, j, p: containsAllergen(p, a) ∧ soldIn(p, j) ∧ CanadaJurisdiction(j) ∧ requiresDeclarationOf(j, a) → mustDeclareInCanada(p, a)
R13	EU declaration duty	∀a, j, p: containsAllergen(p, a) ∧ soldIn(p, j) ∧ EUJurisdiction(j) ∧ requiresDeclarationOf(j, a) → mustDeclareInEU(p, a)
R14	Gluten-free threshold exceeded	∀c, p, x: hasLabelClaim(p, c) ∧ GlutenFreeClaim(c) ∧ hasGlutenPPM(p, x) ∧ x ≥ 20.0 → contradictsClaim(p, c)
R15	Gluten-free threshold met	∀p, x: Product(p) ∧ hasGlutenPPM(p, x) ∧ x < 20.0 → MeetsGlutenFreeThreshold(p)
R16	Sulphites at or above 10 ppm	∀p, x: Product(p) ∧ hasSulphiteMgPerKg(p, x) ∧ x ≥ 10.0 → SulphiteAtOrAbove10ppm(p)
R17	Sulphites above 10 mg/kg	∀p, x: Product(p) ∧ hasSulphiteMgPerKg(p, x) ∧ x > 10.0 → SulphiteAbove10ppm(p)

Note. Rules R14 to R17 use SWRL comparison built-ins and are therefore executed by Pellet but not by HermiT.